

Moein Razavi, Ph.D.

Senior AI/ML Engineer · GenAI for Complex Engineering Systems · Mechanical Engineer

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Professional Summary

Senior AI/ML Engineer with a rare dual foundation: B.S. Mechanical Engineering (controls, robotics, FEA, mechatronics) and **Ph.D. with 4.0 GPA** (Texas A&M) — purpose-built for applying **GenAI to complex engineering systems**. **5+ years** architecting production GenAI platforms: **LLM-orchestrated multi-agent pipelines** (LangGraph, RAG, vector DBs), **transformer-based probabilistic forecasting**, and **Monte Carlo decision engines** that act under uncertainty. At **Ford**, delivered a real-time **V2G aggregator/dispatch system** — a grid-scale cyber-physical decision platform — plus multi-agent GenAI systems serving **10 US states** and **10+ TB** of telemetry. Fluent in physics, simulation, and controls on one side; LLMs, agents, and MLOps on the other. Published researcher (**16 papers, 470+ citations, h-index 10**).

Technical Skills

Agentic AI & Multi-Agent Systems: LangGraph · CrewAI · AutoGen · OpenAI Agents SDK · MCP · LLM Routing & Tool Use · Planner/Executor & Supervisor/Worker patterns · ReAct · Self-Reflection · LangSmith Tracing & Eval

Generative AI, LLMs & RAG: GPT-4/4o · Claude · Gemini · Llama · LangChain · Fine-Tuning (LoRA/QLoRA) · Guardrails · Structured Outputs · Tool Calling · FAISS · ChromaDB · Pinecone · Hybrid Retrieval & Re-rankers · Text-to-SQL

Engineering & Physical Systems: FEA (ANSYS, Abaqus) · Multiphysics (COMSOL) · CFD (ANSYS CFX) · CAD (CATIA, SolidWorks) · Control Systems & Model-Based Design (MATLAB/Simulink) · Robotics & Mechatronics (ADAMS, PLC, Arduino, ARM) · MEMS · Biosignal/EEG Processing

Decision Systems & Forecasting: Probabilistic Forecasting (Chronos-2, iTransformer, TFT, Autoformer) · Monte Carlo Simulation · P50/P90 Uncertainty Quantification · Real-Time Dispatch & Optimization · Bayesian Inference · Backtesting

ML / DL & MLOps: PyTorch · TensorFlow · Hugging Face · scikit-learn · XGBoost · Optuna · SHAP · Distributed GPU (Vertex AI) · Airflow · Docker · Kubernetes · Terraform · CI/CD (Tekton, GitHub Actions)

Cloud, Data & Backend: GCP (Vertex AI, BigQuery, Cloud Run) · AWS · Azure OpenAI · PySpark · PostgreSQL · MongoDB · Python (Expert) · TypeScript · SQL · C++ · MATLAB · FastAPI · REST/WebSocket/gRPC · React/Next.js

Professional Experience

Senior Data Scientist & AI/ML Engineer

Aug 2023 – Present

Ford Motor Company Detroit, MI

- **Built real-time V2G (Vehicle-to-Grid) aggregator and dispatch system** — a grid-scale cyber-physical platform combining **probabilistic forecasting (Chronos-2, iTransformer)** with **Monte Carlo simulation (~3,000 scenarios per dispatch)** to optimize bidding and dispatch under uncertainty; **P50/P90 quantile reporting**, sub-second decisions, streaming telemetry feedback
- **Architected production Multi-Agent GenAI Chatbot** (LangGraph, LangChain) with an **LLM Router/Decider layer** dispatching queries across specialized agents — **SQL Agent** (text-to-SQL over BigQuery), **RAG Agent** (FAISS/ChromaDB, hybrid retrieval, re-ranking), **Document Agent** — serving EV analytics in **10 US states**, cutting query latency **35%**
- **Designed multi-agent ingestion platform** (Vertex AI Gemini, LangGraph) with **Intake, Schema Discovery, Extraction, Validation**, and **Orchestrator** agents — a universal, self-healing pipeline replacing **~500 lines** of utility-specific parsers for CSV/Excel/PDF/XML/HTML ingestion
- **Engineered transformer forecasting pipeline** (TFT, Autoformer, Informer) with Optuna search on distributed Vertex AI GPU — EV load forecasts up **10%**, training time down **40%**; feeds rate optimization, dispatch, and capacity planning
- **Built XGBoost decision model** for EV energy bill prediction over telematics, tariffs, and home consumption (**$R^2 = 0.85$**); deployed as real-time inference service with backtesting and drift monitoring
- **Engineered 10+ TB EV telemetry pipelines** (PySpark, BigQuery, Airflow) powering all agents and models; automated KPI reporting for **500+ stakeholders**, shrinking turnaround **45%**
- **Deployed production MLOps infrastructure** on GCP with Tekton, Terraform IaC, and SonarQube — **99.9% uptime**, ingestion time down **25%**

Jun 2023 – Sep 2024

Software Engineer & AI/ML Researcher

CereVu Medical, Inc. San Francisco Bay Area, CA

- **Architected hardware-to-cloud real-time biosignal AI platform** streaming **9 biomarkers** from wearable sensor patches (Nordic UART BLE) through Python microservices to clinician apps with embedded **ROPA pain decision algorithm** — monitoring precision up **30%**
- **Deployed LSTM/CNN models on-device** (Core ML) for real-time edge inference on wearables across **400+ patients** in 5 clinical studies — **15% accuracy lift**

- **Built Python microservices** (FastAPI, WebSocket) handling **10K+ concurrent biosignal streams** (40% latency cut); shipped a React/Next.js clinical dashboard and 2 production iOS apps with zero-downtime CI/CD

Sep 2018 – May 2024

Research Assistant & ML Engineer

Texas A&M University College Station, TX

- **Designed dual-branch transformer** (1D-CNN tokenizer, RoPE, cross-attention, focal loss) for real-time stress detection on **noisy, naturalistic wearable sensor data** (256 participants, 1.7B rows); **AUC 0.993, F1 0.965**, on-device at **60% faster inference**
- **Built OpenSync** — open-source platform synchronizing EEG, eye-tracking, and biosensor streams over WebSocket; adopted by **3 research labs** (**31 citations**, *J. Neuroscience Methods*)
- **Deployed real-time computer vision system** for COVID-19 distancing/mask detection, **98% accuracy** across **50K+ daily interactions** (76 citations)

2016 – 2018

Research Assistant & Engineering Team Lead, MEMS Neural Microelectrodes

Cognitive Sciences & Technologies Council Tehran, Iran

- **Led the engineering team** designing single/planar/3D neural microelectrodes: structural **FEA in ANSYS** cross-validated in **COMSOL Multiphysics**; published at **MEMS 2018**; taught micro/nano-fabrication processes at Amirkabir University
- **Designed and implemented an EEG-based brain-computer interface for robot motion control** (B.S. thesis) — signal acquisition, filtering, feature extraction (MATLAB DSP, EEGLab), classification, and embedded motor control (ARM/Arduino): the full sense-decide-actuate loop of a physical system

Jan 2023 – Present

Founder & Lead Developer

Clink Social | Inspecalytics McKinney, TX

- **Built and shipped two production platforms solo**: a Firebase social app with real-time messaging and secure QR cross-device OAuth login (engagement up **22%**), and a Supabase field-inspection suite with offline-first sync and in-app DOCX/PPTX report generation (report cycles down **35%**)

Education

Ph.D., Industrial Engineering (Data Science) | GPA: 4.0/4.0 | Research Fellowship

Aug 2021 – Dec 2023

Texas A&M University, College Station, TX

Dissertation: Advancing Stress Detection — Real-Time, Naturalistic, and Personalized AI Approaches.

M.S., Computer Science | GPA: 4.0/4.0 | Research Fellowship

Aug 2019 – May 2021

Texas A&M University, College Station, TX

Coursework: Deep Learning, Computer Vision, NLP, Reinforcement Learning, Algorithm Design.

B.S., Mechanical Engineering | Top 1% Nationwide

2013 – 2018

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

Thesis: EEG-based brain-computer interface for robot motion control · Ranked top 1% of ~250,000 in Iran's National University Entrance Exam · Automatic Control, Robotics, FEM, Vibrations, Heat Transfer, CAD/CAM.

Selected Publications (16 papers · 470+ citations · h-index 10)

- Razavi, M.**, et al. "ML & DL for Stress & Mental Disorder Detection: Scoping Review." *JMIR Mental Health*, 2024. **[79 cites]**
Kazeminasab, S., Razavi, M., et al. "Mapping & Navigation in In-Pipe Robots: A Review." *IEEE Access*, 2021. **[85 cites]**
Razavi, M., et al. "Monitoring of Distance & Mask Compliance in COVID-19." *SN Computer Science*, 2022. **[76 cites]**
Razavi, M., et al. "OpenSync: Synchronizing Measures in Neuroscience." *J. Neuroscience Methods*, 2022. **[31 cites]**
Razavi, M., Nadafi, R. "Conceptual Design of Single/Planar/3D Microelectrodes (ANSYS/COMSOL)." *MEMS 2018*.

Certifications & Honors

Generative AI with LLMs – AWS · **Deep Learning Specialization** – DeepLearning.AI · **Machine Learning** – Stanford · **NLP Specialization** – DeepLearning.AI · **Advanced MLOps** – Coursera · **Bayesian Statistics** – Coursera · **Graduate Research Fellowship**, Texas A&M (2019–2023)